

Impact of Texting-Induced Distraction on Driving Behavior Based on Field Operation Tests

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Abstract

Driver distraction has proven to be among the top contributory factors in road crashes worldwide. The involvement of drivers in secondary activities, such as phone usage, conversation with co-passengers, reaching for objects, eating, and so forth, have been shown to be significant causes of in-vehicle driver distraction. Text messaging using mobile phones while driving can be detrimental to the primary driving task as it invokes manual, visual, and cognitive distractions. Numerous studies in the past have analyzed the effect of texting on driver distraction using driving simulators. This study followed a field operation test-based approach to simulate real road conditions and evaluate the impact of texting on driving behavior. The study was conducted at the connected autonomous vehicle testbed at the Indian Institute of Technology Hyderabad, using an instrumented test vehicle. Data on gaze parameters and vehicle kinematics were recorded using eye-tracker glasses and a high-end global positioning system (GPS) data logger. Thirty-one participants drove on the pre-defined path at the testbed under base-line and texting conditions. The findings demonstrated that the texting task significantly affected fixation count, mean speed, and lateral acceleration. The texting task led to a reduction in mean fixation counts, mean speed, and mean lateral acceleration by 44.37%, 34%, and 46.72%, respectively. The age and driving experience of the participants had a significant effect on their behavior. The results indicate the critical potential of texting-based driver distraction and suggest the enforcement of stringent rules for texting while driving to create a safer road environment.

Keywords

driver behavior, distraction, driver performance, texting, cell phone use, human factors, gaze behavior

Road crashes pose a significant global challenge, requiring prompt action and ingenious remedies. The exponential rise in vehicular mobility has resulted in a distressing upsurge in road crashes around the world. Driver distraction is one of the major causes of road crashes. In 2021, a significant number of fatalities in road crashes in the United States, around 3,522, were directly linked to driver distraction (1). These incidents involved at least one driver engaging in distracting activities, leading to the road crash. This statistic accounted for approximately 8.2 % of the recorded fatalities in road crashes reported throughout the year in the United States (1). In India, about 1.6 % of the total road crashes in 2021 were attributed to distraction resulting from mobile phone usage (2).

Driver distractions can be classified into three primary categories: visual, manual, and cognitive (3). Visual distractions encompass activities that divert the driver's

visual attention away from the road, such as using mobile phones or other visually engaging tasks. Manual distractions involve actions that necessitate the driver to physically remove their hands from the steering wheel, like reaching for objects or eating or drinking. Whereas cognitive distractions involve mental engagement in tasks unrelated to driving, such as daydreaming or recollecting memories. These distractions significantly elevate the

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probability of road crashes, highlighting the need to study the impact of secondary task performance while driving.

A range of methodologies have been employed to accurately assess the magnitude and implications of driver distraction on driving performance. One commonly used approach involves the analysis of eye-tracking data and gaze patterns to gauge the impact of distractions on drivers (4). Additionally, physiological indicators, such as heart rate variability and electroencephalography (EEG) signals, serve as valuable metrics for measuring the cognitive workload and stress levels induced by distractions during driving (5). Devices such as cameras, sensors, and data loggers also offer an alternative means of capturing driver distraction activities (6). Furthermore, self-reported questionnaires have proven effective in elucidating drivers' subjective perceptions of situations involving distracted driving (7). These multidimensional measurement techniques offer a thorough framework for evaluating the effects of driver distraction on road safety.

Driver distraction-based road crashes may be reduced, if not eliminated, with the advancements in automation in the automotive sector. Advanced driver assistance systems (ADAS) and autonomous vehicles (AV) are examples of automated driving systems that are effective in reducing human error, improving vehicle control, and enhancing road safety (8, 9). Although the advent of fully autonomous vehicles in the traffic stream might theoretically eliminate road crashes caused by human error, this is far from reality because of the relatively high cost of AVs and inadequate infrastructure in developing economies such as India.

The successful implementation of any new technology-based solution relies greatly on public perception and opinion. Therefore, it is essential to carefully consider this aspect before integrating any solution into existing systems. Questionnaire surveys are regarded as highly effective practical tools given their ability to collect data from a large sample of respondents. The responses obtained can be used further for detailed analysis, providing insights into how road users perceive driver distractions (7).

Although capable of yielding substantial responses, questionnaire surveys are inadequate substitutes for field operation tests (FOT) when conducting comprehensive analyses of driver behavior in real-world driving scenarios. FOTs employ in-vehicle data collection devices to monitor various driver and vehicle-related parameters. Although driving simulators are frequently used in modern research to simulate real-world driving situations (10–13), they fall short of replicating the complexities of real road conditions. Consequently, conducting studies on real roads becomes essential. However, certain critical

distractions, such as cell phone usage while driving, present substantial risks in real traffic conditions. To address this, carefully designed controlled road environments that mirror realistic road geometry and traffic scenarios offer a reliable platform for investigating the effects of distractions on driver performance and safety. This study is one such attempt to understand driving behavior in a realistic road environment.

Through this study, we aim to improve the overall understanding of driver distraction and its underlying causes, with a focus on examining the impacts of texting while driving using the FOT approach.

Literature Review

Performing secondary tasks such as phone calls and texting while driving has been found to significantly impair driver performance, resulting in reduced speeds, increased reaction times, loss of control of the vehicle, and increased road crashes. Researchers in the past have investigated the impact of these secondary tasks while driving using various approaches, as outlined in the following sections.

Surveys

Foreman et al. conducted a discrete choice experiment to assess texting while driving, with 345 adult participants in the study. The participants were given scenarios and asked if they would text and drive in specific circumstances. According to the findings, individuals were more inclined to text when driving on rural roads, and if the person receiving the message was considered important by the driver (14). Moore and Brown assessed the relationship between self-regulation, habitual text messaging, trait mindfulness, and texting while driving. An online survey was incorporated to collect data from 170 respondents. The results showed that habitual texting behavior was significantly associated with higher frequencies of texting while driving. The findings suggested that the drivers texted while driving out of pure habit, which was influenced by deficiencies in self-regulation (15). Hill et al. conducted an online survey-based investigation to examine the attitudes of college students toward distracted driving. The survey attracted 4,964 responses, with the age of the respondents being 21.82 ± 2.95 (mean $\pm$ SD) years. The findings demonstrated that around 90% of the students used mobile phones for phone calls and texting while driving. Moreover, the students expressed the view that stringent regulations on mobile phone usage while driving would influence their behavior (16). Mikoski et al., in their study involving 492 participants aged from 18 to 76 years, examined drivers'

assessments of distraction risk and other driving scenarios. The study revealed a pronounced inclination toward phone use while driving among young drivers compared with their older counterparts. Moreover, the drivers reported feeling more at ease while driving under daylight conditions than when driving at night (7). In an online survey-based study conducted by Oscar Oviedo involving 484 participants, it was found that experienced drivers who considered texting while driving to be a high-risk task and those who supported the concept of reducing driving speeds during phone usage displayed a decreased tendency to engage in texting while driving (17).

Naturalistic Driving Studies (NDS)

Morgenstern et al. investigated whether drivers reduce their speeds while texting and driving on highways. A 134-sample naturalistic driving data set was considered to assess the speeds of the drivers on highways. The results indicated that drivers reduced their speeds by more than 2 km/h while texting and regained their initial speeds after the texting event (18). Schneiderei et al. conducted a naturalistic driving-based study with 131 participant drivers aged from 16 to 74 years. The study's objective was to examine the driving speeds before and after performing secondary tasks, which included texting, eating, and radio adjustment. The analysis revealed the tendency of the drivers to alter their speeds, particularly when driving at high speeds and engaging in secondary tasks (19). Furthermore, Atwood et al. (20) observed that the texting rate while driving had a significant variation based on age, where drivers aged from 16 to 19 years exhibited a texting rate four times greater than the drivers aged from 30 to 64 years. Wang et al. investigated the effect of phone usage on driving performance, focusing on parameters such as speed, lane offset, and headway. The study involved 51 participant drivers aged from 25 to 59 years. The findings demonstrated a significant reduction in the standard deviations of the speed, lane offset, and headway when drivers engaged in mobile phones under naturalistic driving conditions (21). Bastos et al. examined mobile phone usage while driving, involving six participant drivers with the age range from 19 to 38 years. The study revealed prolonged mobile phone use among drivers during longer trips and at lower driving speeds (22).

Christoph et al. examined the interaction times between drivers, mobile phones, and GPS devices. The study comprised 21 drivers with a mean age of 37.3 ± 9.9 (mean $\pm$ SD) years and driving experience of over 10,000 km per year. Results indicated that the drivers spent around 1% and 4% of their driving time engaging with GPS and mobile phones, respectively.

Additionally, it was observed that the drivers maintained their driving speeds while performing visual-manual tasks (23). Simons-Morton et al. examined the visual distraction of drivers away from the road when performing secondary tasks while driving. Forty-two young drivers aged 16.4 ± 0.22 (mean $\pm$ SD) years and holding a driving license for less than three weeks participated in the study. It was observed that longer glances away from the road increased the likelihood of a road crash and near-crash events (24).

Driving Simulator Studies

Driving simulators have been used extensively to study the effect of in-vehicle distractions on driving behavior. Choudhary and Velaga conducted a driving simulator-based study where the authors investigated and modeled the impacts of conversation and texting on the driving performance of Indian drivers in mean speed and crash-avoiding abilities. The results showed that the drivers compensated for the additional workload by reducing their mean speeds by 5.29 m/s during the texting task. Also, phone calls while driving were found to be more dangerous for young drivers because of their risk-taking behavior and lack of experience (10, 25). Yannis et al. investigated the impact of texting on the behavior and safety of teenage drivers on urban and rural roads. The study included 34 drivers aged from 18 to 28 years. The impact of texting while driving was studied, considering parameters such as traffic environment, weather conditions, driver characteristics, self-reported driving behavior, and road crash occurrence. The findings from their study revealed that texting while driving significantly increased the occurrence of road crashes, with a higher likelihood of occurrence of road crashes in urban regions than in rural areas (26). Peng et al. conducted a study on 28 drivers across four age groups, aiming to assess the effect of entering and reading text on in-vehicle information systems (IVIS) while driving. The authors observed that even short text entry tasks could significantly degrade the driver's longitudinal and lateral control, suggesting that time taken for the text entry or reading tasks significantly affected the distraction levels (27). In a study by Drews et al., 40 young drivers aged from 19 to 23 years participated in a high-fidelity driving simulator-based study, performing a single task (driving) and a dual task (driving and text messaging). Findings showed that the participants in the dual-task condition responded to the onset of braking lights with delay and displayed impairments in forward and lateral control compared with participants in the driving-only condition. Additionally, it was observed that drivers who texted while driving had a higher crash rate than non-texting drivers (28). Bendak, in their simulator-based study, assessed the effects of

texting while driving. Twenty-one young male drivers participated in the study, with four different driving scenarios on highways and town roads. Four cameras were fixed to the front and back of the simulator to record driving activities and eye movements. The results revealed five times more road crashes while texting and occurrences of unnecessary lane departure events compared with the drive-only conditions (29).

Field Tests

Field tests on closed controlled tracks have also been used in the past to examine the effects of texting while driving. These studies are safe and effective in simulating distraction-inducing tasks. In a closed-track field experiment conducted by Yager et al., it was observed that texting while driving significantly affected response time and lane deviation. Also, Text message reading and writing rates were much lower while driving than in the office (30). Yager, in another study, examined the impact of the type of text entry, considering manual texting, using Siri, and using Vlingo. It was observed that driver response time was significantly affected by all three text entry methods, and the drivers' feedback revealed that they felt less safe while texting and driving compared with the baseline conditions (31). Owens et al. (32) conducted a field experiment with 20 participant drivers and found that handheld texting while driving induced a high mental workload and significantly affected the visual attention of drivers. Neurauter et al., found that middle-aged drivers had higher lane deviation and speed variance while texting compared with the younger and older drivers. The texting task also significantly increased the drivers' eyes-off-road time compared with the speech-based tasks (33). Table 1 summarizes the existing research on the effect of secondary tasks on driver performance considering the methodologies of investigation discussed above.

Eye-tracker technology plays a crucial role in driver distraction investigations as it enables the observation and analysis of eye movements while engaging in secondary tasks. These devices record and monitor the driver's gaze behavior, providing valuable information about the driver's attention allocation on the road. Najar et al. conducted a study to examine how in-vehicle objects and spatial distractions affect driver gaze behavior and performance, using an FOT. Their research revealed that object-based distraction led to decreased fixation durations and fixation counts on the area of interest (AOI), along with a significant increase in driving errors (4). In another eye-tracking-based study involving 30 drivers, Desmet et al. found that performing a secondary task, such as a phone call while driving, led to fewer fixations on the road signs, vehicles, and the speedometer. Additionally, the visual scan pattern displayed a wider

spatial distribution during the phone call compared with the no-phone call condition (43). Similarly, Hashash et al., in their study with 26 university students, found that using mobile phones for texting and social media led to decreased visual attention to the road, with the texting task exhibiting a more severe impact on driver performance (44). Therefore, eye tracking presents itself as an effective method for quantifying driver distraction levels during secondary task engagement.

As discussed above, numerous studies in the past have explored the impact of mobile phone usage such as texting and phone calls while driving, the majority of which have been conducted using driving simulators. The use of driving simulators offers distinct advantages, such as examination of the driver's behavior across diverse driving scenarios that are difficult to replicate in real-world contexts. This is particularly advantageous in scenarios that involve potential injury, such as the sudden appearance of obstacles on the road. Although driving simulators offer controlled environments and the ability to simulate diverse scenarios, they lack the complexities and uncertainties of real-world driving that field studies can address effectively. Furthermore, considering that the drivers participating in simulator-based studies realize the virtual set-up, they may not engage with the driving scenarios as seriously as they would when driving in real life, even in the absence of traffic.

Researchers have turned to naturalistic driving studies (NDS) to overcome the virtual set-up shortcomings of driving simulator-based studies, as an NDS provides authentic road conditions for investigating driver behavior. An NDS enables drivers to navigate freely, and the data recording devices installed in the vehicles provide the researchers with real-world driving data. Despite their potential benefits, NDS's are characterized by high costs and time demands and are ethically sensitive, offering limited control to the researchers. Moreover, identifying the precise cause of a driver's visual distraction such as traffic signals, pedestrians, mobile phones, animals, passengers, and so forth would require integrating eye-tracking technology into the NDS.

Performing field operations tests in controlled road environments ensures precise testing conditions, allowing for the safe performance of secondary tasks while driving. Despite its advantage of real-world testing, this approach has seen limited adoption in the past for studying driver behavior given the high costs, logistical challenges, and the inability to simulate diverse risky driving scenarios as effectively as driving simulators.

Considering the advantages and drawbacks outlined for these types of study, this current study adopted an FOT-based approach in the absence of traffic, considering the available resources. Additionally, incorporating an eye-tracker device during the study facilitated the

Table 1. Summary of Existing Research on Secondary Task Performance While Driving

Study	Participants						Findings
	Secondary tasks	Study type	Category	Sample size	Age range (years)	Driver experience	Driving performance parameters
Peng et al. (2014) (27)	Text reading and text entry tasks while driving	Driving simulator study	Age groups	7	21.3 ± 0.5 (mean ± SD) 32.4 ± 5.4 46.9 ± 4.6 59.7 ± 3.0 21.6 ± 3.67 (mean ± SD)	NA	Lane departure, standard deviation of lane positioning, time headway (mean and SD)
He et al. (2014) (34)	Texting (speech and handheld) while driving	Driving simulator study	NA	35	≥ 2 years	mean = 3.5 years	Speed, lane position, brake reaction time
Yannis et al. (2014) (26)	Texting while driving	Driving simulator study	NA	34	18–28	mean = 3.5 years	Speed, reaction time, road crash probability
Rumschlag et al. (2015) (11)	Texting while driving	Driving simulator study	6–10 texts/week 11–50 texts/week 51–500 texts/week >500 texts/week	6 15 22 7	34.5 ± 11.7 (mean ± SD)	18 ± 12 (mean ± SD) years	Number of lane excursions, percent of texting time in lane excursions
Choudhary, P. and N.R. Velaga (2017) (35)	Texting and conversation while driving	Driving simulator study	Young Middle age Older age	55 39 6	< 30 30 ≤ 50 ≥ 50	mean = 4.5 years mean = 11.92 years mean = 34.33 years	Lateral acceleration, steering reversal rate, steering wheel angle, number of lane excursions, standard deviation of lane positioning
Xu, J. and Y. Lin. (2018) (36)	Texting while driving	Driving simulator study	NA	36	18–56	NA	Speed, average headway, platoon length, reaction time
Choudhary, P. and N.R. Velaga (2018) (25)	Texting and conversation while driving	Driving simulator study	In-experienced young drivers	24	< 25	< 4 years	Speed, longitudinal acceleration, steering reversal rate, standard deviation of lane positioning
Young et al. (2018) (37)	Texting while driving	Driving simulator study	Professional drivers NA	25 28	≥ 25 18–55	> 5 years Average = 9.8 h per week	Brake reaction time, mean speed, standard deviation of lateral position, eye glance on/off the road
Vollrath et al. (2021) (38)	Texting while driving	Driving simulator study	NA	40	28 ± 12.6 (mean ± SD)	Average = 10 years	Mean deviation from lane
Choudhary et al. (2022) (39)	Texting, conversation, operating music player, eating	Driving simulator study	NA	90	≤ 30	> 1 year	Reaction time, mean speed, standard deviation of lane positioning
Christoph et al. (2013) (23)	Interaction with mobile phone and GPS	Naturalistic driving study	NA	21	37.3 ± 9.9 (mean ± SD)	> 10,000 km/year	Interaction time with mobile phone/GPS device

(continued)

Table 1. (continued)

Study	Participants					Driving performance parameters		Findings
	Secondary tasks	Study type	Category	Sample size	Age range (years)	Driver experience	Driving performance parameters	
Simons-Morton et al. (2014) (24)	Secondary tasks during driving	Naturalistic driving study	NA	42	16.4 ± 0.22 (mean ± SD)	Holding a driving license for 3 weeks or less	Total duration of eye glance and single longest eye glance off the road	High road crash risk was observed for the single longest glance during a secondary task in comparison to the total duration of eye glance off the road.
Schneider et al. (2017) (19)	Texting while driving	Naturalistic driving study	NA	131	16–74	0–50 years	Speed	Drivers tend to adjust speeds when performing a secondary task while driving.
Christoph et al. (2019) (40)	Phone usage while driving	Naturalistic driving study	NA	28	44.5 ± 12.9 (mean ± SD)	NA	Mobile phone usage time and frequency	Drivers spent 9% of their driving time engaging with mobile phones, and the presence of a passenger was associated with a reduction in mobile phone usage by the drivers.
Wang et al. (2020) (21)	Phone usage while driving	Naturalistic driving study	NA	51	25–59	NA	Speed, lane offset, headway	Significant reductions in standard deviations of speed, lane offset, and headway were observed during phone usage while driving.
Morgenstern et al. (2020) (18)	Texting while driving	Naturalistic driving study	NA	34	38 ± 11.30 (mean ± SD)	NA	Speed	Drivers significantly reduce their driving speeds when texting while driving.
Bastos et al. (2020) (22)	Phone usage while driving	Naturalistic driving study	NA	6	19–38	0–21 years	Mobile phone usage time, mobile phone usage frequency, speed	Drivers displayed reduced involvement in complex tasks on mobile phones, such as texting, whereas in general mobile phone usage increased during longer trips and at lower driving speeds.
Hill et al. (2015) (16)	Driver distractions	Survey	NA	4,964	21.82 ± 2.95 (mean ± SD)	NA	Demographic information, perception on distracted driving	About 90% of students reported being engaged in phone calls and texting while driving. Regular drivers experienced frequent distractions. Moreover, students believed that regulations against phone usage while driving would influence their driving behavior.
Oviedo-Trespalacios, O. (2018) (17)	Texting/browsing while driving	Survey	NA	162	17–65	NA	Demographic information, attitude toward texting/browsing while driving	Experienced drivers who perceived texting while driving as high risk and supported reducing speeds during phone usage demonstrated a decreased likelihood of texting/browsing while driving.
Mikoski et al. (2019) (41)	Driver distractions	Survey	NA	492	18–76	NA	Demographic information, perception on distraction risk, poor visibility conditions, safety behavior	Young drivers had a higher frequency of phone usage while driving than older drivers. Drivers admitted using mobile phones even after knowing the distraction risks involved.
Hill et al. (2019) (42)	Texting while driving	Survey	NA	220	19–70	NA	Demographic information, driving-based variables, hands-free device usage	Dependence on mobile phones for everyday activity and social factors are major contributors leading to texting while driving.
Moore, M.M. and P.M. Brown (2019) (15)	Texting while driving	Survey	NA	170	28.11 ± 12.04 (mean ± SD)	NA	Demographic information, questions on mobile phone use for texting	Texting while driving is a result of pure habit. Additionally, the tendency of drivers to text while driving depends on their level of trait mindfulness.

(continued)

Table 1. (continued)

Study	Participants					Driving performance parameters	Findings
	Secondary tasks	Study type	Category	Sample size	Age range (years)	Driver experience	
Owens et al. (2010) (32)	Texting (handheld and in-vehicle systems)	Field test (closed track)	Female (19–26 years) Female (40–50 years) Male (19–34 years) Male (39–51 years)	5 4 6 5	24 (mean) 44 (mean) 27 (mean) 48 (mean)	NA	Using handheld phones to send and receive text messages induced higher mental demand and interfered with drivers' visual and steering behaviors. The older drivers were more affected by the texting tasks compared with the younger drivers. Texting while driving significantly affected light detection response times. The drivers' lane-keeping ability, assessed through lane position deviation, was notably affected by texting while driving. The text message reading and writing rates decreased significantly when driving compared with performing the same tasks in the office.
Yager et al. (2012) (30)	Writing and reading text messages	Field test (closed track)	Male (n = 17) Male (n = 25)	42	16–54	NA	Handheld texting success rates were higher among younger (96%) compared with middle-aged (82%) participants. Middle-aged drivers showed higher lane deviation and speed variance during handheld texting than younger drivers. On average, middle-aged drivers had lane deviations for 34% of the texting session, compared with 18% for younger drivers. Handheld texting significantly increased eyes-off-road time (68%) compared with speech-based tasks (26%). Text messaging tasks significantly affected drivers' response times across three conditions: manual entry, using Siri, and using Vlingo. The response times were delayed by a factor of 1.92 for manual texting, 1.87 for Siri, and 1.77 for Vlingo.
Neurauter et al. (2012) (33)	Handheld texting and speech-based texting approach	Field test (closed track)	Young drivers Middle-aged drivers	24	18–30 45–55	NA	In the 'send only' task, there was no significant difference between the three conditions. However, in the three 'read and reply' tasks, manual texting performed the best, while Siri performed the worst. Additionally, participants believed their driving improved when using Siri for texting compared with manual texting or Vlingo.
Yager, C.E. (2013) (31)	Texting (manually, using Siri, Vlingo)	Field test (closed track)	NA	43	16–63	NA	Text messaging tasks significantly affected drivers' response times across three conditions: manual entry, using Siri, and using Vlingo. The response times were delayed by a factor of 1.92 for manual texting, 1.87 for Siri, and 1.77 for Vlingo.

Note: SD = standard deviation; GPS = global positioning system; NA = Not available.

examination of the cause of visual distraction. This approach would offer several advantages: a) enhanced control over the desired driving scenarios under examination; b) significant cost efficiency; c) ethical alignment; and d) mitigation of possible hazardous conditions that could compromise safety. The deliberate exclusion of road traffic from the study was justified, considering that texting while driving in real-world traffic, especially in countries such as India characterized by mixed traffic conditions, would involve high complexity and extreme risk. Moreover, the absence of prior research that investigated the pure distraction potential of texting while driving under realistic road and weather conditions in the absence of traffic highlights the novelty of this study. Consequently, the current research displays the potential to serve as a pilot investigation in this domain, offering valuable insights into the sole distraction potential of texting while driving.

The principal objective of this research is to assess the exclusive impact of texting on driver performance under a realistic road environment, focusing on quantifiable metrics, which include speed, acceleration, and gaze parameters such as fixation count and duration.

Methodology

Participants

Convenience and snowball sampling techniques were incorporated to recruit participants for the study. The majority of the participants were students and the working staff of the Indian Institute of Technology Hyderabad. Participation in the study was voluntary, and the participants could withdraw from the study at any stage. The advertisement for participation in the study was circulated by e-mail. The shortlisting criteria for participation in the study included possession of a valid Indian driving license (manual transmission car) and a minimum of one year of driving experience in real-world traffic conditions. Thirty-three participants voluntarily participated in the study. Data from two participants were considered invalid because of poor instrument calibration and data not being recorded appropriately. The final study sample comprised of 31 drivers (29 males, 2 females) aged from 19 to 42 years ($M = 25.54$, $SD = 4.08$), with a mean driving experience of 4.93 years. The participants received incentives in the form of gifts as a token of appreciation at the end of the study. The purpose of the study was clearly expressed to the participants, and the participants signed a written consent form before the start of the study.

Instrumentation

The FOT was conducted using an instrumented passenger car equipped with four video cameras, a GPS device,

ADAS, and a Vbox pro data logger (Figure 1). Also, the drivers wore the Pupil's invisible eye-tracker glasses during the study drive. The GPS device was mounted on top of the car and was used to get the vehicle's coordinates during the study. The four video cameras were set up inside the car to give a clear view of the outer road environment and the driver. The data logger was used to record the kinematic parameters, such as the speed and acceleration of the car during the drive. Pupil's invisible eye-tracker glasses were used to record the gaze movement of the driver during the study (Figure 2) (45).

Test Route

Since the test involved driving under distracted conditions, a controlled road environment set-up was used for the study. The Technology Innovation Hub on Autonomous Navigation and Data Acquisition Systems (TiHAN) connected autonomous vehicles (CAV) testbed facility at the Indian Institute of Technology (IIT) Hyderabad was used as a controlled road environment for the study. The testbed allowed for the testing of critical scenarios that could occur in real traffic conditions. The testbed facility was equipped with varying road geometry, including multilane roads, curved road sections, S-curves, and intersections. Also, traffic control devices such as traffic signs, signals, and road markings were available at the testbed for efficient use. The total road network at the testbed facility was approximately 1 km in length (Figure 3).

Procedure

Figure 4 depicts the brief experimental procedure considered in the study. A detailed briefing was given to the participants before the start of the study to familiarize them with the test conditions. Calibration of the eye-tracker device was done before the trial drive. Also, adjustments for powered lenses were considered if the participant wore eyeglasses with powered lenses. Before beginning the study drive, the participants drove on the testbed to get accustomed to the test route and the instrumentation inside the vehicle. The driver's seat and mirrors were adjusted to fit each driver's comfort. All participants signed a written consent form with details related to the conduct of the study, risks involved, benefits, data security, and so forth. Once the participants felt confident and comfortable in using the instrumented test vehicle for the study, they were taken to the starting point of the test route at the testbed.

During the study drive, the participants drove twice on the defined path. First under the baseline condition (no distraction) and the second time under the text messaging condition. During the second trip, the participants

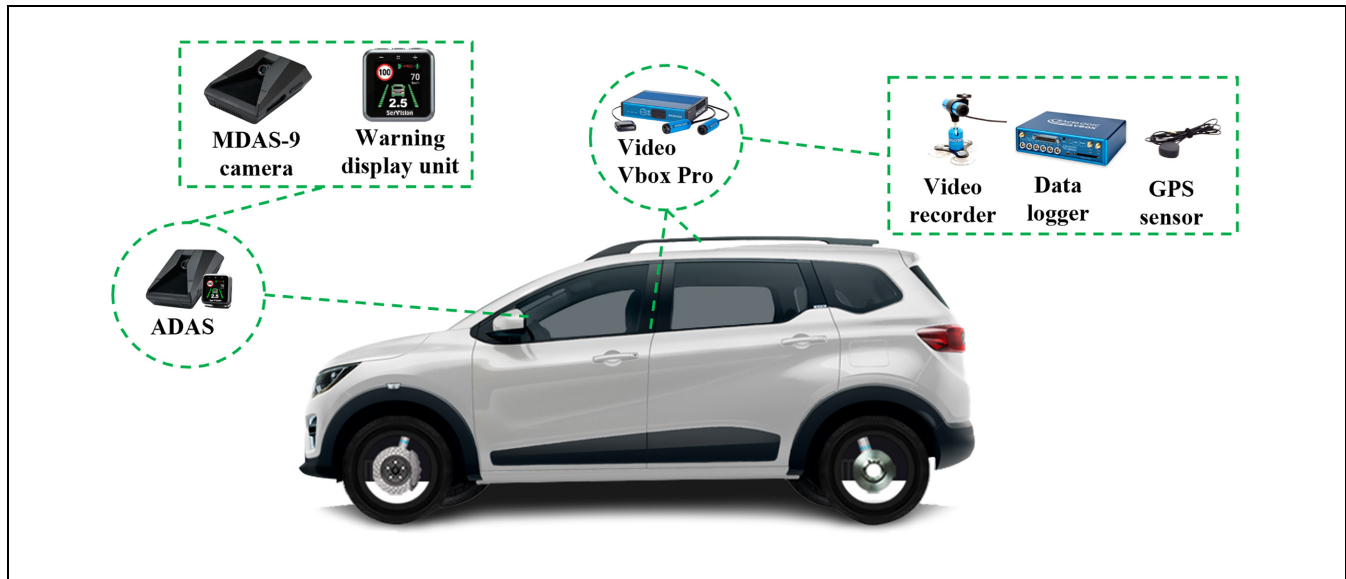


Figure 1. Instrumented passenger car.

Note: MDAS = Movon driver assistance system; ADAS = advanced driver assistance systems; GPS = global positioning system.

drove while being engaged in reading and replying to text messages as the car was in motion. The text messages included questions such as “What is your favorite food dish?” “Write down the title of your favorite song.”, and “What is your favorite color?” The participants were instructed not to reply to the text messages when they stopped at any intersection during a red-light signal. A Samsung M21 2021 edition smartphone was used to complete the texting task, with a QWERTY keyboard setting and type/swipe options enabled. A post-experiment questionnaire was shared with the participants after completing the driving-based study. The questionnaire comprised various sections for providing feedback on the instrumentation and collecting participant demographic characteristics. A view of the FOT conducted during the study is shown in Figure 5.

Results

A significance level (α) of 0.05 was considered for statistical inference throughout this study. The effect of texting on driving performance was studied by considering the following parameters: fixation count, fixation duration, mean speed, longitudinal acceleration, and lateral acceleration. Driving experience and age were considered as classification parameters for the driving participants to analyze if age and driving experience influenced driving performance in a distracted state caused by the texting task. The participant’s age and driving experience data were collected through the post-experiment questionnaire. Based on age, the drivers were classified into two age groups: a) less than or equal to 25 years, and b)

more than 25 years. Also, based on their driving experience, the drivers were classified into three groups: a) less than 2 years, b) 2 to 5 years, and c) more than 5 years.

The data sets considered for analysis were assessed for parametric properties, such as meeting the assumptions of normality and homogeneity of variance, which are necessary for parametric statistical analysis. If the data were found to satisfy these assumptions, parametric statistical tests were employed; otherwise, non-parametric statistical tests were used for analysis. In the rest of this paper, “parametric in nature” refers to data sets that satisfied the parametric assumptions, whereas “non-parametric in nature” indicates data sets that did not meet these assumptions.

Gaze Behavior

In the study, gaze was defined as the driver’s point of observation inside and outside the car. Meanwhile, fixation was represented as the eye movement fixed at any object of interest. The gaze behavior of the driver was analyzed for the fixation count and fixation duration on the AOI considered in the study. Fixation count denoted the number of fixations on the AOI, whereas fixation duration denoted the time duration for which the eye remained fixated on the AOI. iMotions software was used to analyze the gaze-related data obtained from the Pupil’s invisible eye-tracker glasses. As shown in Figure 6, the AOIs considered were the roadway in front and the mobile phone used for text messaging during the study. The fixation count and duration on the roadway under the baseline and texting conditions were compared and tested for significant differences.

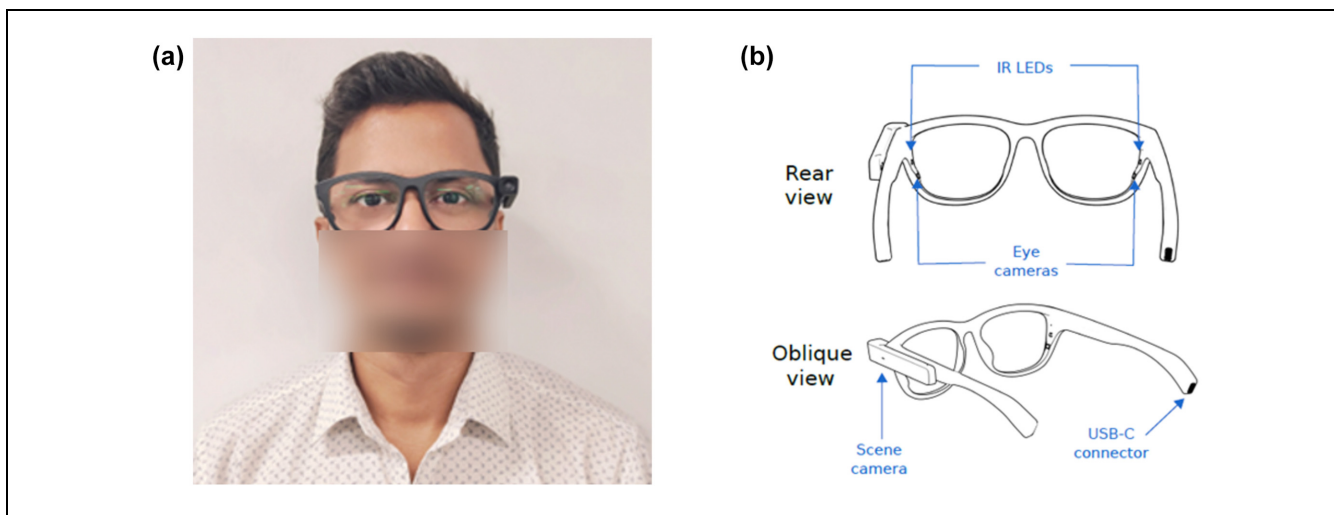


Figure 2. Pupil invisible eye tracker: (a) device in position (photo taken and owned by the first author); and (b) line diagram of eye-tracker device (45).

Fixation Count

A Wilcoxon paired test was used to check for significant differences between fixation counts on the road, under baseline and texting conditions, as the data were non-parametric in nature. A significant difference in the fixation counts was observed for the driving conditions, with lesser fixations on the road during the texting condition ($p < 0.05$, Cohen's $d = 1.219$) (Figure 7).

Effect of Driving Experience and Age on Fixation Count. A paired T-test was used to check for significant differences between the same driving experience group when driving under varying conditions (baseline and texting), as the data were parametric in nature. A significant difference in the mean fixation count was observed for all three groups: “less than 2 years” ($p = 0.001$, Cohen's $d = 2.36$), “2 to 5 years” ($p = 0.003$, Cohen's $d = 0.867$), and “more than 5 years” ($p < 0.05$, Cohen's $d = 0.704$) (Figure 8). T-tests with Bonferroni adjustments were used to check for significant differences among the driving experience groups under the same driving conditions. No significant differences were observed under the baseline and texting conditions.

A Wilcoxon paired test was used to check for significant differences between the same age group when driving under varying conditions (baseline and texting), as the data were non-parametric in nature. A significant difference in the mean fixation count was observed in both the age groups: “less than or equal to 25 years” ($p < 0.05$, Cohen's $d = 0.907$), and “more than 25 years” ($p < 0.05$, Cohen's $d = 1.744$) (Figure 9). A Wilcoxon test was used to check for significant differences between age groups under the same driving conditions. No significant

differences were observed under the baseline condition ($p = 0.09$) or the texting condition ($p = 0.66$).

Fixation Duration

The mean fixation duration data were found to be parametric in nature, therefore a paired T-test was used to check for significant differences in fixation durations when driving under baseline and texting conditions. No significant difference was observed for fixation durations under the two driving conditions ($p = 0.889$) (Figure 10).

Effect of Driving Experience and Age on Fixation Duration. A paired T-test was used to test significant differences between the same driving experience group when driving under varying conditions (baseline and texting), as the data were parametric in nature. There was no significant difference observed in the mean fixation duration for all three groups: “less than 2 years” ($p = 0.07$), “2 to 5 years” ($p = 0.97$), and “more than 5 years” ($p = 0.32$) (Figure 11). T-tests with Bonferroni adjustments were used to check for significant differences among the driving experience groups under the same driving conditions. No significant differences were observed under the baseline and texting conditions.

Paired T-tests were considered to check for significant differences between the same age group when driving under varying conditions (baseline and texting), as the data were parametric in nature. There was no significant difference observed in the mean fixation durations for the two age groups: “less than or equal to 25 years” ($p = 0.16$) and “more than 25 years” ($p = 0.08$) (Figure 12). Unpaired T-test was used to check for significant differences between age groups under the same

driving conditions. No significant differences were observed under baseline ($p = 0.4$) and texting conditions ($p = 0.2$).

Mean Speed

The vehicle kinematic parameters such as speed and acceleration were obtained from the Vbox pro data logger, and analyses were conducted to determine if texting

while driving had any effect on the mean speed of the driver during the study.

The mean speed data were normally distributed, but the equal variance condition was not satisfied. Therefore, Welch's test was used to check for significant differences in mean speeds when driving under baseline and texting conditions. A significant difference was observed for mean speeds under the two driving conditions ($p < 0.05$, Cohen's $d = 1.782$) (Figure 13). The probable reasons for the drivers reducing their speeds while texting could be the distracting nature of the texting task, prioritization of safety concerns, difficulty in multitasking, limited peripheral vision when focusing on the phone, and reduced confidence levels to drive without continuously observing the road.

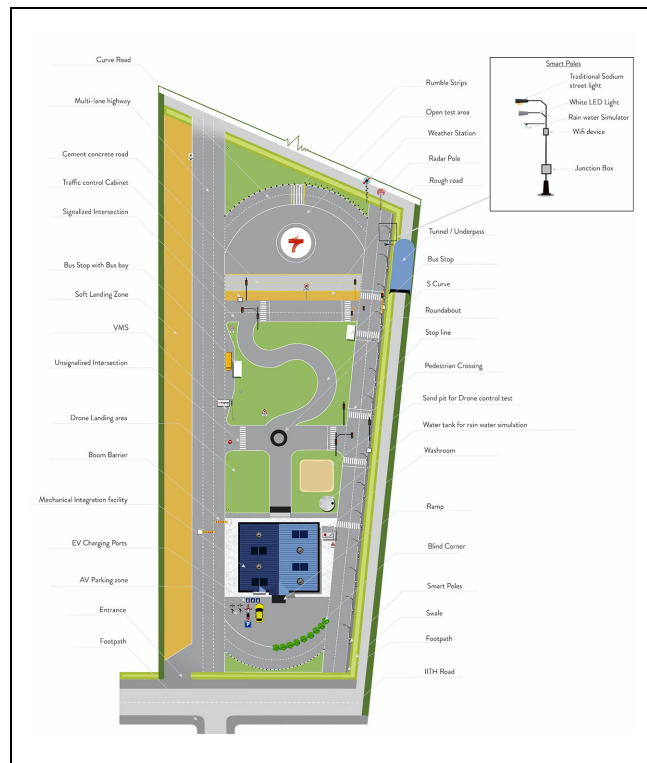


Figure 3. CAV testbed at IIT Hyderabad.

Note: CAV = connected autonomous vehicle; IIT = Indian Institute of Technology; VMS = Variable message sign; EV = electric vehicle; AV = autonomous vehicle.

Effect of Driving Experience and Age on Speed. A paired T-test was used to test significant differences between the same driving experience group when driving under varying conditions (baseline and texting), as the data were parametric in nature. Significant differences were observed in the mean speeds for all three groups: “less than 2 years” ($p = 0.001$, Cohen's $d = 1.366$), “2 to 5 years” ($p = 0.003$, Cohen's $d = 1.98$), and “more than 5 years” ($p < 0.05$, Cohen's $d = 2.495$) (Figure 14). T-tests with Bonferroni adjustments were used to check for significant differences among the driving experience groups under the same driving condition. No significant differences were observed under the baseline and texting conditions.

Paired T-tests were considered to check for significant differences between the same age group when driving under varying conditions (baseline and texting), as the data were parametric in nature. Significant differences were observed in the mean speeds for the two age groups: “less than or equal to 25 years” ($p < 0.05$, Cohen's $d = 1.63$), and “more than 25 years” ($p < 0.05$, Cohen's $d = 1.94$) (Figure 15). An unpaired T-test was used to

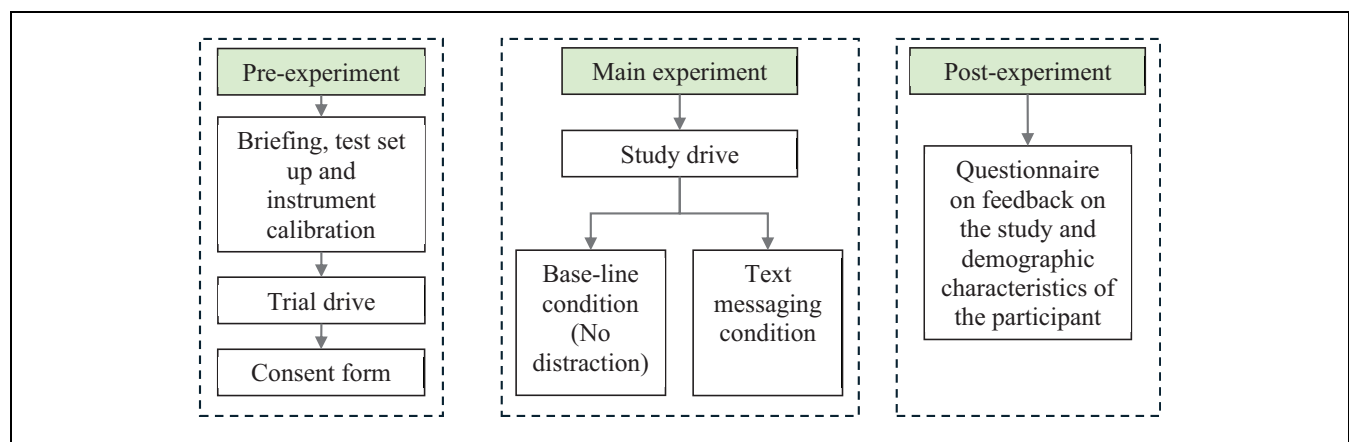


Figure 4. Experimental procedure.



Figure 5. FOT study using instrumented passenger car at CAV testbed IIT Hyderabad (photo taken and owned by the first author).

Note: FOT = field operation test; CAV = connected autonomous vehicle; IIT = Indian Institute of Technology.

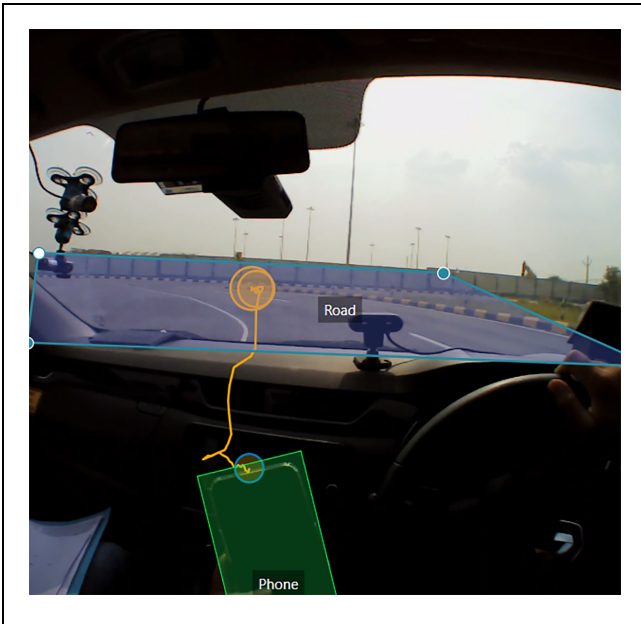


Figure 6. Gaze analysis using iMotions software.

check for significant differences between age groups under the same driving conditions. No significant differences were observed under baseline condition ($p = 0.73$) and texting condition ($p = 0.81$).

Acceleration

The data logger set-up in the instrumented test vehicle provided the vehicle's longitudinal and lateral acceleration data for analysis. Longitudinal acceleration refers to the acceleration or deceleration experienced by the vehicle along the longitudinal axis and is associated with changes in vehicle speeds. On the other hand, lateral

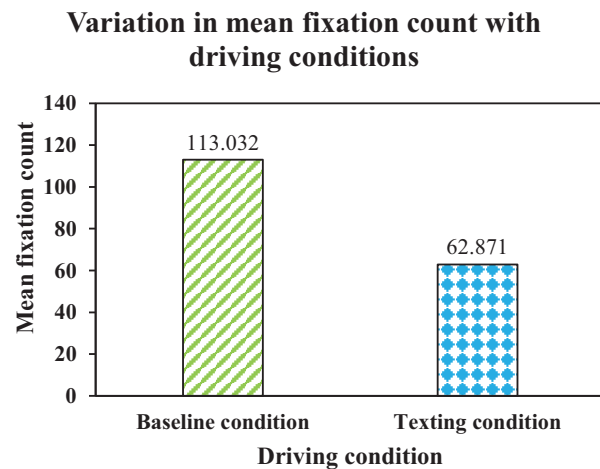


Figure 7. Variation in mean fixation count with driving conditions.

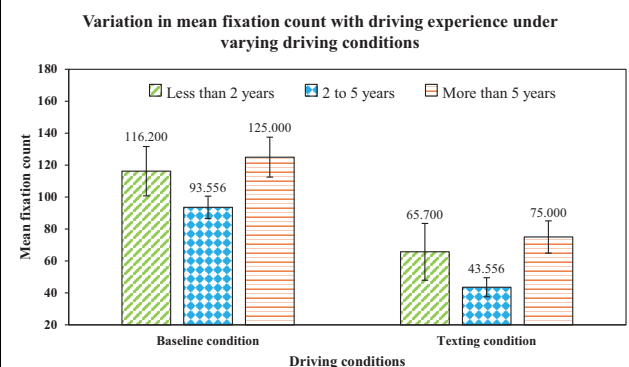


Figure 8. Variation in mean fixation count with driving experience under varying driving conditions, where error bars represent $\pm$ one SEM.

Note: SEM = standard error of the mean.

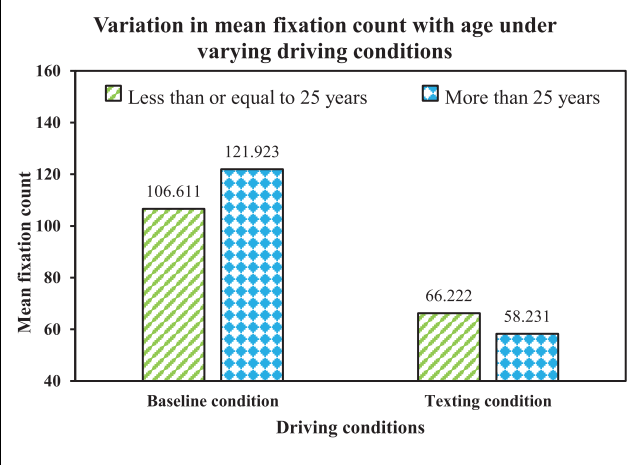


Figure 9. Variation in mean fixation count with age under varying driving conditions.

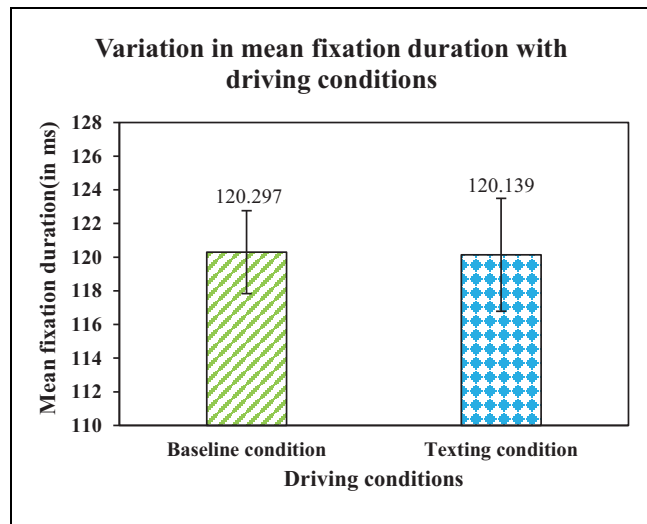


Figure 10. Variation in mean fixation duration with driving conditions, where error bars represent $\pm$ one SEM.
Note: SEM = standard error of the mean.

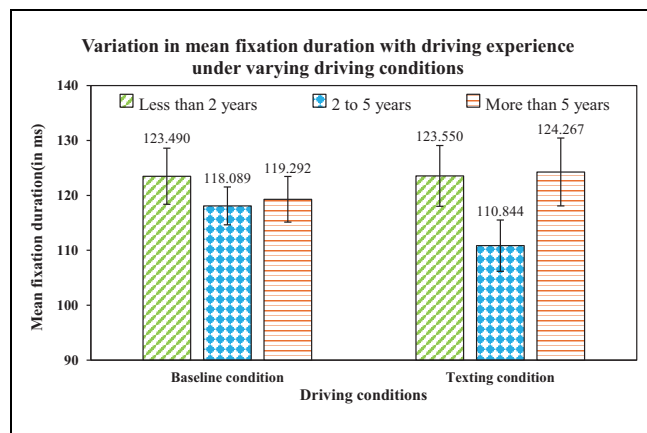


Figure 11. Variation in mean fixation duration with driving experience under varying driving conditions, where error bars represent $\pm$ one SEM.
Note: SEM = standard error of the mean.

acceleration is the acceleration experienced in the side-ways direction perpendicular to the longitudinal axis. It may be considered as a parameter for measuring the vehicle's stability and is critical in preventing skid and slip occurrences.

Longitudinal Acceleration

The mean longitudinal acceleration data were observed to be normally distributed, but the equal variance condition was not satisfied. Therefore, Welch's test was used to check for significant differences in mean longitudinal acceleration when driving under baseline and texting conditions. No significant difference was observed for mean longitudinal acceleration under the two driving conditions ($p = 0.460$) (Figure 16).

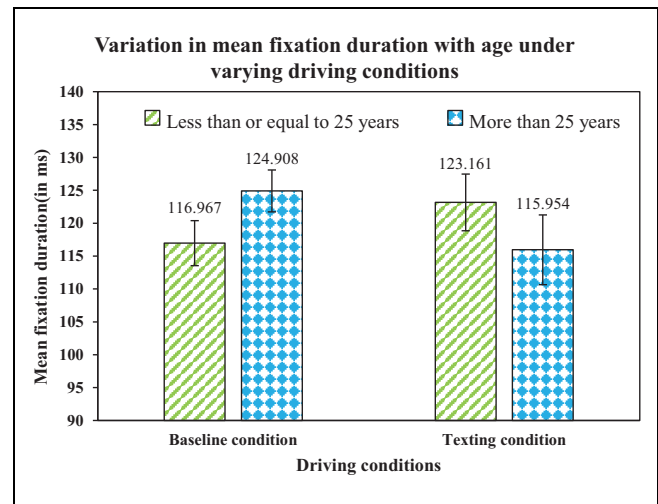


Figure 12. Variation in mean fixation duration with age under varying driving conditions, where error bars represent $\pm$ one SEM.
Note: SEM = standard error of the mean.

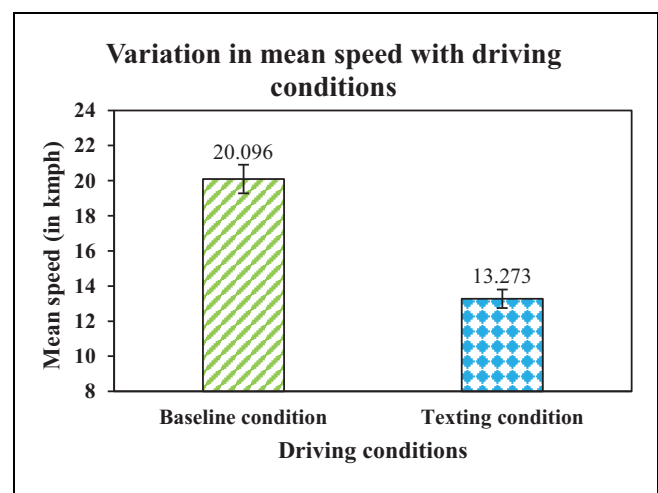


Figure 13. Variation in mean speed with driving conditions, where error bars represent $\pm$ one SEM.
Note: SEM = standard error of the mean.

Effect of Driving Experience and Age on Longitudinal Acceleration. A paired T-test was used to check for significant differences between the same driving experience group when driving under varying conditions (baseline and texting), as the data were parametric in nature. No significant differences were observed in the mean longitudinal accelerations for all three groups: "less than 2 years" ($p = 0.07$), "2 to 5 years" ($p = 0.79$), and "more than 5 years" ($p = 0.47$) (Figure 17). T-tests with Bonferroni adjustments were used to check for significant differences among the driving experience groups under the same driving conditions. No significant differences in mean longitudinal acceleration were observed under the baseline and texting conditions.

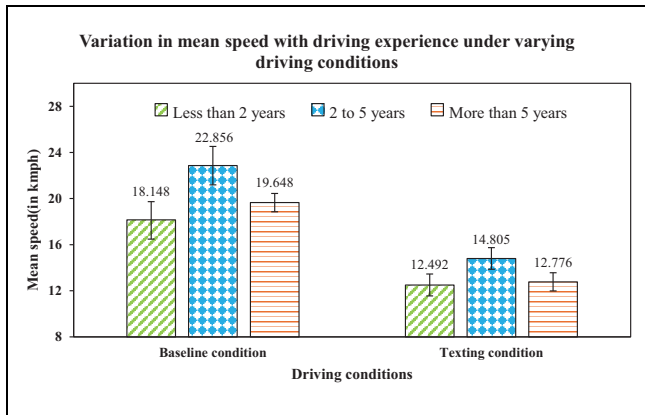


Figure 14. Variation in mean speed with driving experience under varying driving conditions, where error bars represent $\pm$ one SEM.

Note: SEM = standard error of the mean.

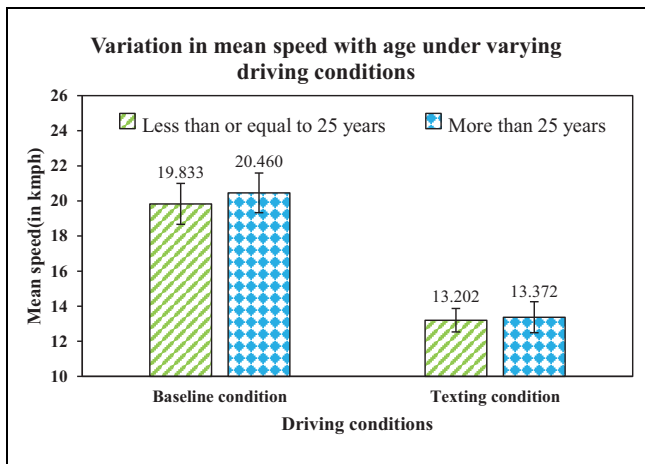


Figure 15. Variation in mean speed with age under varying driving conditions, where error bars represent $\pm$ one SEM.

Note: SEM = standard error of the mean.

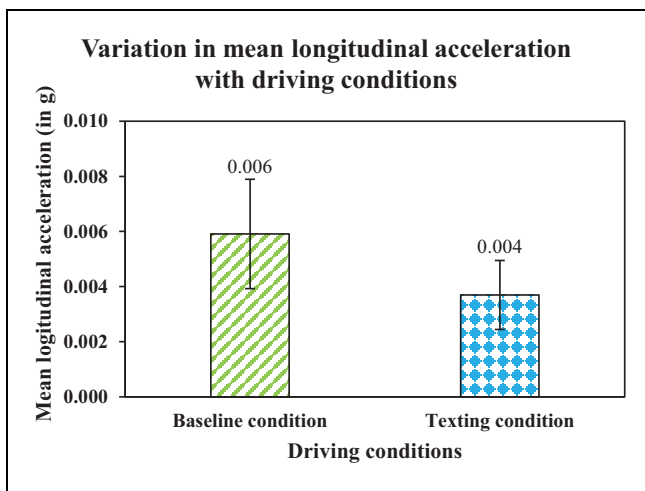


Figure 16. Variation in mean longitudinal acceleration with driving conditions, where error bars represent $\pm$ one SEM.

Note: SEM = standard error of the mean.

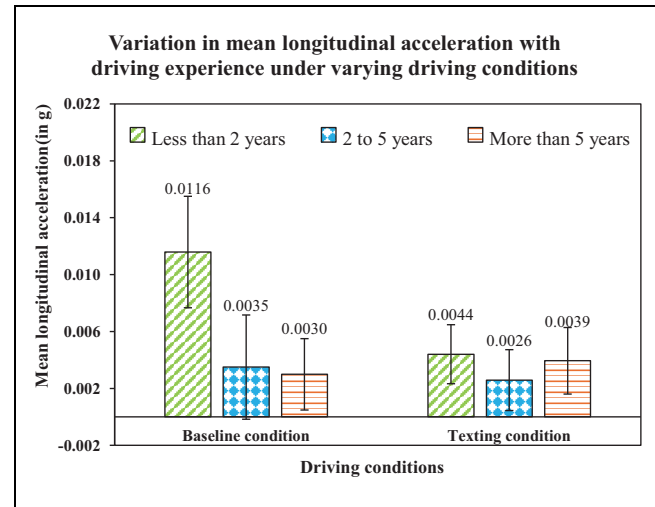


Figure 17. Variation in mean longitudinal acceleration with driving experience under varying driving conditions, where error bars represent $\pm$ one SEM.

Note: SEM = standard error of the mean.

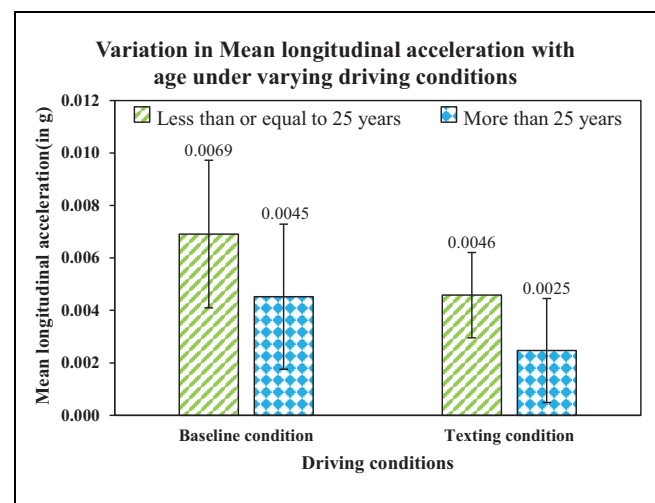


Figure 18. Variation in mean longitudinal acceleration with age under varying driving conditions, where error bars represent $\pm$ one SEM.

Note: SEM = standard error of the mean.

Paired T-tests were considered to test for significant differences between the same age group when driving under varying conditions (baseline and texting), as the data were found to be normally distributed. No significant difference was observed in the mean longitudinal acceleration for the two age groups: “less than or equal to 25 years” ($p = 0.69$) and “more than 25 years” ($p = 0.47$) (Figure 18). Welch’s test was used to check for significant differences between age groups under the same driving conditions. No significant differences were observed under the baseline condition ($p = 0.09$) and the texting condition ($p = 0.23$).

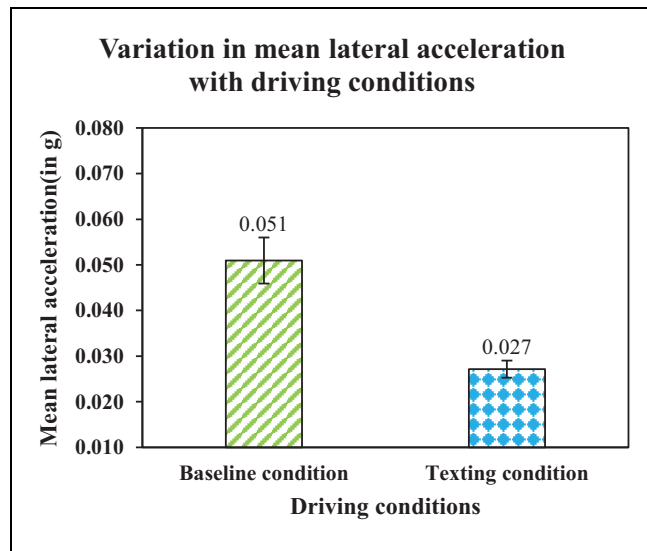


Figure 19. Variation in mean lateral acceleration with driving conditions, where error bars represent $\pm$ one SEM.
Note: SEM = standard error of the mean.

Lateral Acceleration

The mean lateral acceleration data were observed to be normally distributed, but the equal variance condition was not satisfied. Therefore, Welch's test was used to check for significant differences in mean lateral acceleration when driving under baseline and texting conditions. A significant difference was observed for mean lateral acceleration under the two driving conditions ($p < 0.05$, Cohen's $d = 1.109$) (Figure 19).

Effect of Driving Experience and Age on Lateral Acceleration. A paired T-test was used to check for significant differences between the same driving experience group when driving under varying conditions (baseline and texting), as the data were parametric in nature. No significant differences were observed in the mean lateral accelerations for two groups: "2 to 5 years" ($p = 0.0519$), and "more than 5 years" ($p = 0.06$), but a significant difference was observed for the group "less than 2 years" ($p < 0.05$, Cohen's $d = 2.36$) (Figure 20). T-tests with Bonferroni adjustments were used to check for significant differences among the driving experience groups under the same driving conditions. No significant differences in mean lateral acceleration were observed under the baseline and texting conditions.

Paired T-tests were considered to check for significant differences between the same age group when driving under varying conditions (baseline and texting), as the data were parametric in nature. A significant difference was observed in the mean lateral acceleration for the two age groups: "less than or equal to 25 years" ($p = 0.007$, Cohen's $d = 0.87$) and "more than 25 years" ($p = 0.0002$,

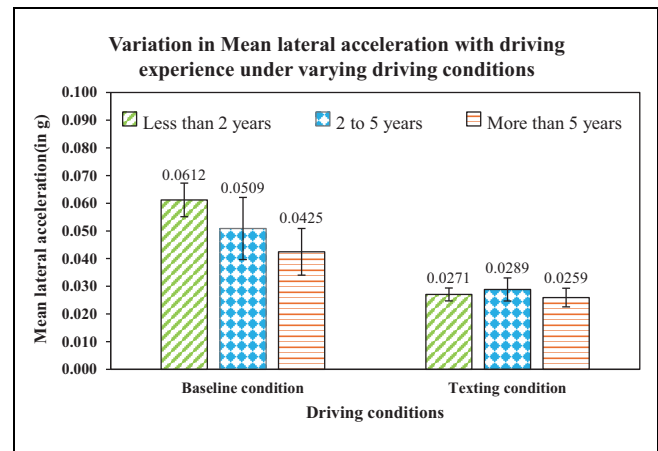


Figure 20. Variation in mean lateral acceleration with driving experience under varying driving conditions, where error bars represent $\pm$ one SEM.
Note: SEM = standard error of the mean.

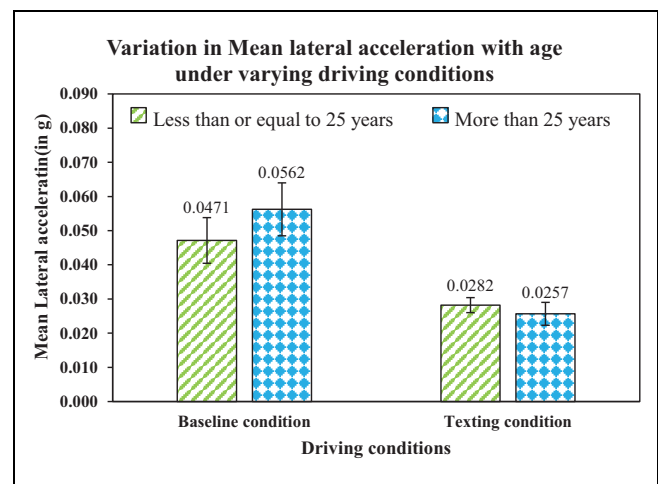


Figure 21. Variation in mean lateral acceleration with age under varying driving conditions, where error bars represent $\pm$ one SEM.
Note: SEM = standard error of the mean.

Cohen's $d = 1.42$) (Figure 21). An unpaired T-test was used to check for significant differences between age groups under the same driving conditions. No significant differences were observed under the baseline condition ($p = 0.08$) and texting condition ($p = 0.78$).

Discussion

The present study aimed to investigate the sole distraction effect of texting while driving on driver behavior in a real road setting. Driver behavior was assessed through the driver gaze and kinematic parameters. The observations are discussed below, focusing on individual parameters sequentially.

Fixation Count

The observations from the analysis highlighted the critical distraction potential of texting as a secondary task, as the mean fixation count on the road was reduced by 44.37% when the participants texted while driving. The reduction in the fixation count on the AOI (roadway) during an in-vehicle distraction task was consistent with the observations recorded by Najar et al. (4). When different driving experience groups were considered for analysis, it was observed that the reduction in mean fixation count on the road was highest for “less than 2 years” group, followed by “2 to 5 years” and “more than 5 years.” This observation indicated that the more experienced drivers would get distracted to a lesser extent during the texting operation, as shown by the mean fixation count. While comparing the age groups for distraction extent, the study revealed that the drivers from the “more than 25 years” age group were distracted to a greater extent as compared with the “less than or equal to 25 years” age group, indicating that the distraction levels were proportional to driver age. A possible explanation for this observation could be the early exposure of the younger age group to mobile phones and other high-end technologies compared with older individuals. Early exposure to mobile phones might have led to habituated mobile phone use while performing other primary tasks.

Fixation Duration

The analysis revealed that the drivers’ mean fixation durations were not significantly affected by changes in driving conditions (baseline and texting conditions). These findings contradicted the observations of the distraction-based study conducted by Najar et al. (4). Classifications of drivers based on driving experience and age also did not bring out any significant difference. A few probable reasons for this could be the habituation of the driver to texting while driving, driving experience under similar conditions, and the relatively small sample size of participants in the study that could not yield significant differences in the mean fixation duration values. Based on the study’s observations, it was inferred that the mean fixation durations of the drivers could be independent of driving conditions, age, and driving experience.

Mean Speed

The results from the study showed that the mean speeds on the road were reduced by 34% (6.82 km/h) when the participants drove under the texting condition. The significant reduction in speeds during the texting task aligned with the observations from previous studies (10,

12, 18). When different driving experience groups were considered for analysis, it was observed that the reduction in mean speed was the highest for the “more than 5 years” group, followed by the “2 to 5 years” and “less than 2 years” groups. This indicated that the driving experience was proportional to the reduction in mean speeds of drivers under texting conditions and that the more experienced drivers tended to prioritize safety. While comparing the age groups for significant changes in mean speed, it was observed that the “more than 25 years” age group had a higher speed differential than the “less than or equal to 25 years” age group. This observation indicated that drivers with greater driving experience and older age prioritized safety and therefore significantly reduced their mean speeds while texting and driving.

Longitudinal Acceleration

It was observed that the mean longitudinal acceleration of drivers was not significantly affected by changes in driving conditions (baseline and texting conditions). Classifications of drivers based on driving experience and age also did not bring out any significant change in longitudinal acceleration values. These observations suggested that longitudinal acceleration may have been among the factors independent of driving conditions and notably dependent on the individual driving behavior.

Lateral Acceleration

The results revealed that the mean lateral acceleration was reduced by 46.72% when the participants drove under texting conditions. The decreased lateral acceleration during the texting condition would result in the reduced ability of the driver to change direction or avoid obstacles swiftly. Moreover, when encountering a sudden obstacle on the road, the driver may oversteer, resulting in overcompensation and potentially destabilizing the vehicle’s trajectory, thus hampering safety. When different driving experience groups were considered for analysis, a significant difference in mean lateral acceleration was only observed for the “less than 2 years” group. These results align with previous studies, which have shown that greater driving experience leads to less variation in lateral acceleration under distracted driving conditions (25). While comparing the age groups for significant changes in mean lateral acceleration, it was observed that the “more than 25 years” age group had a higher difference in mean lateral acceleration than the “less than or equal to 25 years” age group. The significant difference in lateral acceleration under distracted driving conditions, with a more pronounced effect in older age groups, aligned with the findings reported in

previous studies (46, 47). The findings of this study highlighted that drivers with less driving experience and higher age were more vulnerable to loss of maneuverability when texting while driving.

Effect Size (Cohen's d)

The results from the analysis revealed that fixation count, mean speed, and lateral acceleration were significantly affected by the distracted condition caused by texting while driving. The Cohen's *d* values observed were 1.219, 1.782, and 1.109 for the fixation count, mean speed, and lateral acceleration, respectively. These values indicated that the mean speed of the driver was most significantly affected, followed by fixation count, and lateral acceleration.

Applications

Texting while driving has the potential to divert the driver's cognitive attention away from the primary task of driving, which highlights the importance of assessing the impact of texting on driver behavior to ensure the safety of all road users. The study was based on an FOT conducted in a controlled road environment, with 31 drivers taking part in the experiment. The study examined the gaze and kinematic parameters to assess the impact of texting on driving behavior. Despite the limitation of the sample size of participants in the study, the results indicated a significant effect of texting on driver behavior. The findings from the study and their possible applications have been outlined below,

Gaze Behavior. The fixation count and duration were used to gauge the visual attention of the driver on the road. The results revealed that texting led to a significant reduction in the fixation count on the road. These findings highlight the risk associated with texting while driving, as eyes off the road is one of the major causes of road crashes. These insights could be used in public safety campaigns and driver education programs to emphasize the importance of visual attention on the road. Also, automotive manufacturers could consider using the findings from the study to develop ADAS that monitor drivers, give warning alerts, and apply corrective measures when visual attention is compromised.

Kinematic Parameters. Speed, longitudinal acceleration, and lateral acceleration were considered as kinematic parameters to assess driver performance during the study. The results showed that texting while driving had a significant impact on the speed and lateral acceleration of the drivers. These findings highlight that texting while driving compromised vehicle control and stability. These

findings could be used by policymakers and government agencies to strengthen laws on mobile phone usage while driving, particularly texting.

Effect of Age and Driving Experience. It was observed that drivers from the less driving experience and higher age groups showed significant differences in the lateral acceleration and fixation count while texting and driving compared with baseline driving conditions. Higher age and greater driving experience groups showed a significantly greater reduction in speeds during the texting condition. This could be considered as a cautious and safety-positive behavior, as lowering speeds while being distracted would lessen the impact of any potential crash. Although the lowering of speeds would, in one way, mitigate the impact of a potential crash, it would still compromise traffic flow and efficiency. Therefore, although experienced drivers perform better under distraction than less experienced ones, texting while driving cannot be justified for any level of experience or age group of drivers. These insights from the study could be used by insurance companies in adjusting premiums based on age and driving experience, considering the risks involved resulting from mobile phone usage while driving.

Overall Impact. Overall, the findings from the study offer valuable insights for policymakers, educational campaigns, industry stakeholders, and government agencies involved in addressing distraction resulting from mobile phone usage while driving, given the profound impact of texting on driver behavior. Notably, the observed decline in visual attention through this study, evidenced by a nearly 45% reduction in fixation count on the roadway, poses significant safety risks. Furthermore, the varying effect of age and driving experience on driver behavior in texting-induced distracted scenarios highlights the importance of tailored regulatory measures. Policymakers and government agencies could thus consider implementing specific guidelines and regulations for different age and experience groups of drivers, thereby enhancing overall traffic efficiency and road safety.

Limitations and Future Scope

Some of the limitations of the current study, which form a base for future studies, are as follows,

- There was no traffic on the roads during the study. As the current study focused on identifying the distraction caused purely by texting conditions, in the absence of any other distraction caused by other potential distraction elements such as other vehicles, pedestrians, or any other object on the roadway.

- The current study only considered text messaging as a distraction-inducing task. Future research could include other secondary tasks while driving such as phone calls, reaching for objects, eating/drinking, driver–passenger conversations, and so forth.
- The sample size considered during the study was relatively small, which could have influenced the outcomes of the results and findings from the study. Also, the age group and gender ratio considered during the study could be improved, as considerably younger participants took part as drivers, with 29 male participants and only two female. Future studies may consider a larger sample size, varied ages, and a balanced gender ratio.

Conclusion

This study was conceptualized to investigate the effects of texting as a secondary task on driving behavior, measured in gaze parameters (fixation count and duration) and vehicle kinematics (speed, longitudinal acceleration, and lateral acceleration). The results found that texting while driving significantly affected the mean speed, lateral acceleration, and fixation count on the roadway. Cohen's *d* values revealed that the texting task had a maximum effect on the mean speed, followed by the fixation count and lateral acceleration. The independent variables considered for the study, such as age and driving experience, significantly affected the driver's behavior. When fixation count was considered, it was observed that higher driving experience led to lower distraction levels, whereas the distraction levels increased with age. When mean speed was considered, it was observed that drivers from the greater experience and older age groups considered in the study reduced their driving speeds to a greater extent as compared with drivers from the less experienced and younger age groups. When considering the lateral acceleration, it was observed that drivers from higher age groups had a greater difference in lateral acceleration than drivers of younger age groups. The study attempted to demonstrate the adverse consequences of text messaging while driving using an FOT approach, as previous studies based on driving simulators suggest that the use of real road driving conditions could yield more precise inferences on the impact of texting while driving (25, 37, 41). As the current study did not consider the road crash risk scenarios, the use of an "FOT approach in a naturalistic road set-up" over a driving simulator may be justified. Also, the findings emphasize the necessity for stringent rules against texting while driving because of the significant safety hazard that it presents to the safety of the driver and other road users.

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Author Contributions

The authors confirm their contribution to the paper as follows: study conception and design: A. Singh, R. Pachamuthu, D. S. Pawar; data collection: A. Singh, A. Srivastava; analysis and interpretation of results: A. Singh, A. Srivastava; draft manuscript preparation: A. Singh, D. S. Pawar. All authors reviewed the results and approved the final version of the manuscript.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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